SMARTBRIDGE
Let's Bridge the Gap

SKILL WALLET
A SmartBridge Product

Project Initialization and Planning Phase

Date	15 March 2024
Team ID	xxxxxx
Project Title	Credit Card Auto-Approval Prediction System
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) Report

The proposal report aims to transform credit card approval using machine learning, boosting efficiency and accuracy. It tackles system inefficiencies, promising better operations, reduced credit risk, and happier customers. Key features include a machine learning-based credit card scoring model and real-time decision-making.

Project Overview	
Objective	• Objective: The primary objective is to automate the credit card application review process by implementing a machine learning classification pipeline that provides instant, objective eligibility decisions.
Scope	• Scope: The project covers the end-to-end development pipeline: analyzing historical applicant demographic and financial dataset profiles, training predictive models, packaging the optimal Random Forest model (95.8% accuracy), and deploying an interactive Flask-based web application.
Problem Statement	
Description	• Description: Manual verification of credit card applications is operationally slow, expensive, and subject to human inconsistency. Outdated scoring frameworks fail to accurately model the non-linear relationships within complex demographic and financial criteria.
Impact	• Impact: Manual bottlenecks lead to extended customer wait times and lost opportunities. Additionally, inaccurate credit scoring either increases bad-debt risk for banks or incorrectly rejects creditworthy applicants.
Proposed Solution	
Approach	• Approach: Constructing a high-accuracy binary classification machine learning pipeline using Python and Scikit-Learn. The pipeline scales numerical values, encodes categorical applicant properties, and predicts creditworthiness to serve instantaneous decisions.

Key Features	• Real-time prediction endpoint returning instant approval/rejection outcomes and confidence probabilities. • Interactive, user-friendly HTML5 input interface customized for applicant financial profiles (income, assets, employment). • Preprocessed Exploratory Data Analysis (EDA) visualizations showing demographic impact trends.

Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	CPU/GPU specifications, number of cores	Apple M-Series (Silicon) / standard Multi-Core Intel CPU
Memory	RAM specifications	8 GB RAM (minimum)
Storage	Disk space for data, models, and logs	256 GB SSD (requires less than 50MB disk space for the serialization model and web runtime)
Software		
Frameworks	Python frameworks	Flask (Python micro-framework)
Libraries	Additional libraries	scikit-learn, pandas, numpy, matplotlib, seaborn
Development Environment	IDE	VS Code / GitHub Codespaces / Jupyter Notebook
Data		
Data	Source, size, format	Credit Card Approval Dataset, CSV format